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An elementary introduction to information geometry

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Abstract

In this survey, we describe the fundamental differential-geometric structures of information manifolds, state the fundamental theorem of information geometry, and illustrate some use cases of these information manifolds in information sciences. The exposition is self-contained by concisely introducing the necessary concepts of differential geometry, but proofs are omitted for brevity.

Keywords: Differential geometry; metric tensor; affine connection; metric compatibility; conjugate connections; dual metric-compatible parallel transport; information manifold; statistical manifold; curvature and flatness; dually flat manifolds; Hessian manifolds; exponential family; mixture family; statistical divergence; parameter divergence; separable divergence; Fisher-Rao distance; statistical invariance; Bayesian hypothesis testing; mixture clustering; αembeddings; gauge freedom

1 Introduction

1.1 Overview of information geometry

We present a concise and modern view of the basic structures lying at the heart of *Information Geometry* (IG), and report some applications of those information-geometric manifolds (herein termed "information manifolds") in statistics (Bayesian hypothesis testing) and machine learning (statistical mixture clustering).

By analogy to *Information Theory* (IT) (pioneered by Claude Shannon in his celebrated 1948's paper [119]) which considers primarily the communication of messages over noisy transmission channels, we may define *Information Sciences* (IS) as the fields that study "communication" between (noisy/imperfect) data and families of models (postulated as *a priori* knowledge). In short, information sciences seek methods to *distill* information from data to models. Thus information sciences encompass information theory but also include the fields of Probability & Statistics, Machine Learning (ML), Artificial Intelligence (AI), Mathematical Programming, just to name a few.

We review some key milestones of information geometry and report some definitions of the field by its pioneers in §5.2. Professor Shun-ichi Amari, the founder of modern information geometry, defined information geometry in the preface of his latest textbook [8] as follows: "Information geometry is a method of exploring the world of information by means of modern geometry." In short, information geometry geometrically investigates information sciences. It is a mathematical endeavour to define and bound the term geometry itself as geometry is open-ended. Often, we start by studying the invariance of a problem (eg., invariance of distance between probability distributions) and get as a result a novel geometric structure (eg., a "statistical manifold"). However, a geometric structure is "pure" and thus may be applied to other application areas beyond the scope of the original problem (eg, use of the dualistic structure of statistical manifolds in mathematical programming [57]): the method of geometry [9] thus yields a pattern of abduction [103, 115].

A narrower definition of information geometry can be stated as the field that studies the *geometry of decision making*. This definition also includes *model fitting* (inference) which can be interpreted as a decision problem as illustrated in Figure 1: Namely, deciding which model parameter to choose from a family of parametric models. This framework was advocated by Abraham Wald [131, 132, 36] who considered all statistical problems as statistical decision problems. Dissimilarities (also loosely called distances among

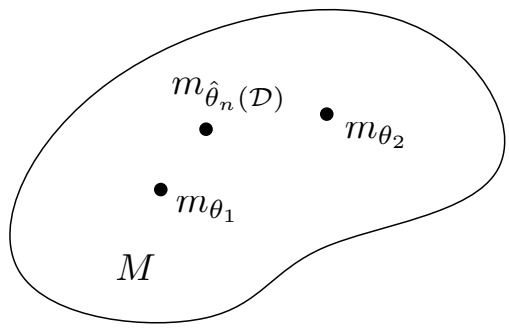

Figure 1: The parameter inference $\hat{\theta}$ of a model from data $\mathcal{D}$ can also be interpreted as a decision making problem: Decide which parameter of a parametric family of models $M = \{m_\theta\}_{\theta\in\Theta}$ suits the "best" the data. Information geometry provides a differential-geometric structure on manifold M which useful for designing and studying statistical decision rules.

others) play a crucial role not only for measuring the *goodness-of-fit* of data to model (say, likelihood in statistics, classifier loss functions in ML, objective functions in mathematical programming or operations research, etc.) but also for measuring the discrepancy (or deviance) between models.

One may ponder why adopting a geometric approach? Geometry allows one to study *invariance* of "figures" in a coordinate-free framework. The *geometric language* (e.g., line, ball or projection) also provides affordances that help us reason intuitively about problems. Note that although figures can be visualized (i.e., plotted in coordinate charts), they should be thought of as purely abstract objects, namely, geometric figures.

Geometry also allows one to study *equivariance*: For example, the centroid $c(T)$ of a triangle is equivariant under any affine transformation A: $c(A.T) = A.c(T)$. In Statistics, the Maximum Likelihood Estimator (MLE) is equivariant under a monotonic transformation g of the model parameter θ: $\widehat{(g(\theta))} = g(\hat{\theta})$, where the MLE of θ is denoted by $\hat{\theta}$.

1.2 Outline of the survey

This survey is organized as follows:

In the first part (§2), we start by concisely introducing the necessary background on differential geometry in order to define a manifold structure (M, g, ∇), ie., a manifold M equipped with a metric tensor field g and an affine connection ∇. We explain how this framework generalizes the Riemannian manifolds (M, g) by stating the fundamental theorem of Riemannian geometry that defines a unique torsion-free metric-compatible Levi-Civita connection which can be derived from the metric tensor.

In the second part (§3), we explain the dualistic structures of information manifolds: We present the conjugate connection manifolds (M, g, ∇, ∇^*), the statistical manifolds (M, g, C) where C denotes a cubic tensor, and show how to derive a family of information manifolds $(M, g, \nabla^{-\alpha}, \nabla^{\alpha})$ for $\alpha \in \mathbb{R}$ provided any given pair $(\nabla = \nabla^{-1}, \nabla^* = \nabla^1)$ of conjugate connections. We explain how to get conjugate connections ∇ and ∇^* coupled to the metric g from any smooth (potentially asymmetric) distances (called divergences), present the dually flat manifolds obtained when considering Bregman divergences, and define, when dealing with parametric family of probability models, the exponential connection ${}^e\nabla$ and the mixture connection ${}^m\nabla$ that are dual connections coupled to the Fisher information metric. We discuss the concept of statistical invariance for the metric tensor and the notion of information monotonicity for statistical divergences [30, 8]. It follows that the Fisher information metric is the unique invariant metric (up to a scaling factor), and that the f-divergences are the unique separable invariant divergences.

In the third part (§4), we illustrate how to use these information-geometric structures in simple applications: First, we described the natural gradient descent method in §4.1 and its relationships with the

Riemannian gradient descent and the Bregman mirror descent. Second, we consider two applications in dually flat spaces in §4.2: In the first application, we consider the problem of Bayesian hypothesis testing and show how Chernoff information (which defines the best error exponent) can be geometrically characterized on the dually flat structure of an exponential family manifold. In the second application, we show how to cluster statistical mixtures sharing the same component distributions on the dually flat mixture family manifold.

Finally, we conclude in §5 by summarizing the important concepts and structures of information geometry, and by providing further references and textbooks [25, 8] for further readings to more advanced structures and applications of information geometry. We also mention recent studies of generic classes of principled distances and divergences.

In the Appendix §A, we show how to estimate the statistical f-divergences between two probability distributions in order to ensure that the estimates are non-negative in §B, and report the canonical decomposition of the multivariate Gaussian family, an example of exponential family which admits a dually flat structure.

At the beginning of each part, we start by outlining its contents. A summary of the notations used throughout this survey is provided page 47.

2 Prerequisite: Basics of differential geometry

In §2.1, we review the very basics of Differential Geometry (DG) for defining a manifold (M, g, ∇) equipped with both a metric tensor field g and an affine connection ∇. We explain these two *independent* metric/connection structures in §2.2 and in §2.3, respectively. From an affine connection ∇, we show how to derive the notion of covariant derivative in §2.3.1, parallel transport in §2.3.2 and geodesics in §2.3.3. We further explain the *intrinsic curvature and torsion* of manifolds induced by the connection in §2.3.4, and state the fundamental theorem of Riemannian geometry in §2.4: The existence of a unique torsion-free Levi-Civita connection ${}^{\mathrm{LC}}\nabla$ compatible with the metric (metric connection) that can be derived from the metric tensor g. Thus the Riemannian geometry (M, g) is obtained as a special case of the more general manifold structure $(M, g, {}^{\mathrm{LC}}\nabla)$: $(M, g) \equiv (M, g, {}^{\mathrm{LC}}\nabla)$. Information geometry shall further consider a dual structure (M, g, ∇^*) associated to (M, g, ∇), and the pair of dual structures shall form an information manifold (M, g, ∇, ∇^*).

2.1 Overview of differential geometry: Manifold (M, g, ∇)

Informally speaking, a *smooth D-dimensional manifold M* is a topological space that locally behaves like the D-dimensional Euclidean space $\mathbb{R}^D$. Geometric objects (e.g., points, balls, and vector fields) and entities (e.g., functions and differential operators) live on M, and are *coordinate-free* but can conveniently be expressed in *any* local coordinate system of an atlas $\mathcal{A} = \{(\mathcal{U}_i, x_i)\}_i$ of charts$(\mathcal{U}_i, x_i)$'s (fully covering the manifold) for calculations. Historically, René Descartes (1596-1650) allegedly invented the global Cartesian coordinate system while wondering how to locate a fly on the ceiling from his bed. In practice, we shall use the most expedient coordinate system to facilitate calculations. In information geometry, we usually handle a single chart fully covering the manifold.

A C^k manifold is obtained when the *change of chart transformations* are C^k. The manifold is said smooth when it is C^∞. At each point $p \in M$, a tangent plane T_p locally best linearizes the manifold. On any smooth manifold M, we can define two <u>*independent*</u> structures:

1. a metric tensor g, and
2. an affine connection ∇.

The metric tensor g induces on each tangent plane T_p an *inner product space* that allows one to measure vector magnitudes (vector "lengths") and angles/orthogonality between vectors. The affine connection ∇ is a differential operator that allows one to define:

1. the *covariant derivative operator* which provides a way to calculate differentials of a vector field Y with respect to another vector field X: Namely, the covariant derivative $\nabla_X Y$,